

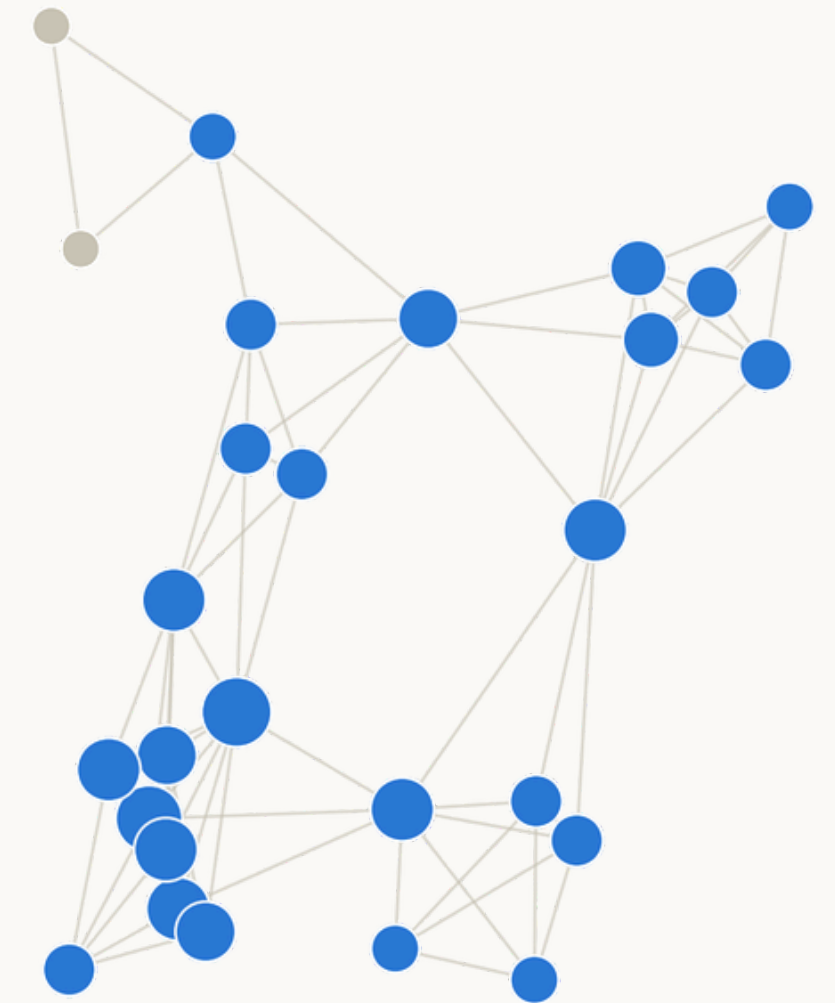
Network Position and Academic Performance

Does where you sit in a friendship network predict your grades, beyond who your direct friends are?

Group members:

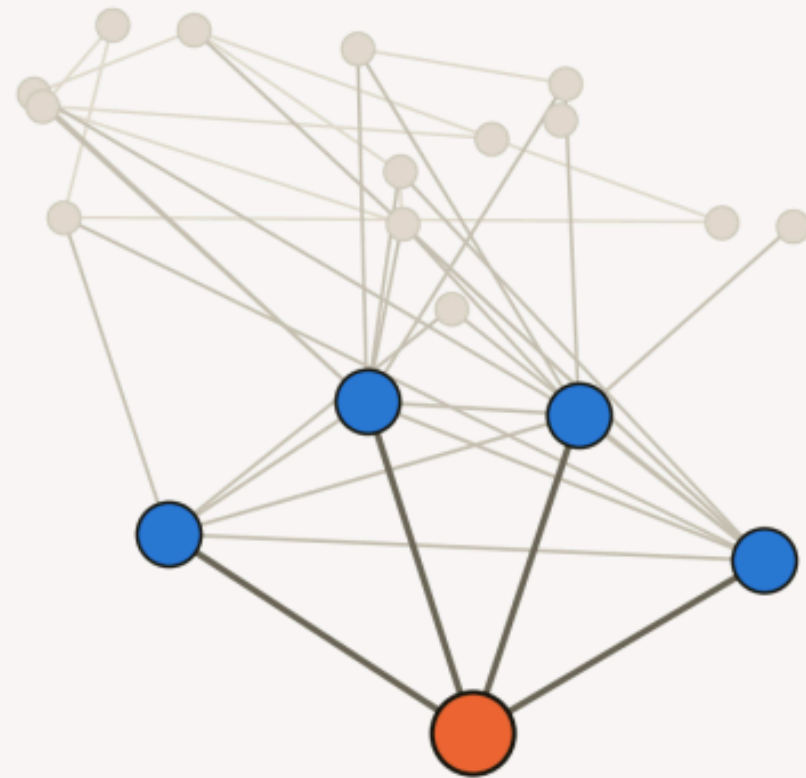
Amy Bing

Tingying Huang

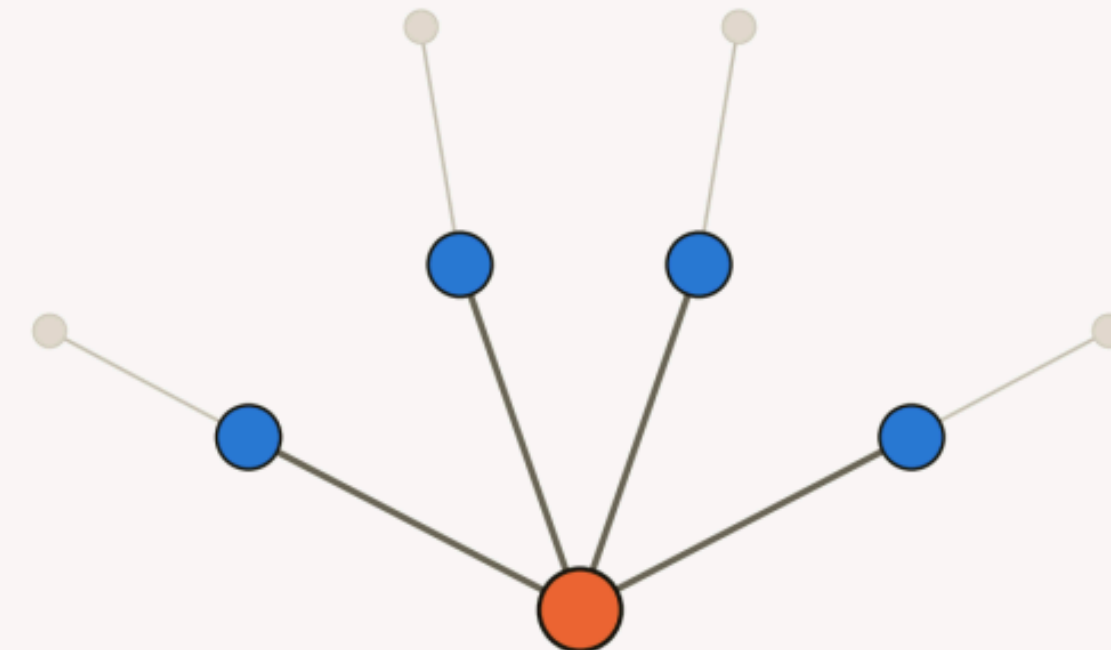


Two students. Same-looking friend groups. Same grades?

Same 4 friends. Same average friend GPA. Same grade?



Student A — 4 friends, gateway to the school



Student B — 4 friends, dead end

Does a student's position in a friendship network affect their grades?

THE DATA

A directed friendship network, observed over time

NODES

655 students in high school

1,200 –1,549 per university year

EDGES

Directed “like” ties, taken from a social media site.

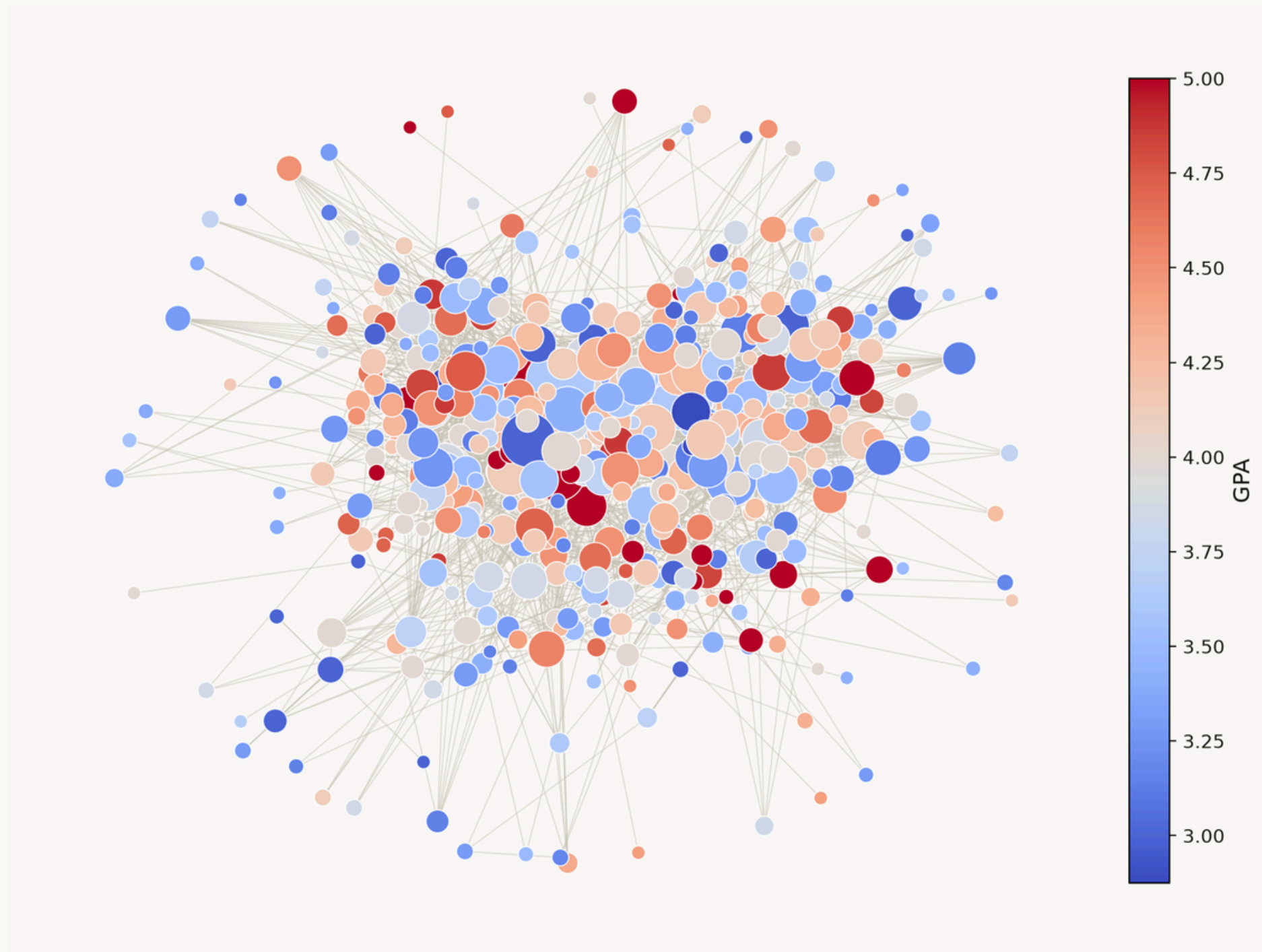
$i \rightarrow j$ means i liked j at least once in a 3-month window.

2–14 snapshots per group, 38 in total.

Only 24–27% are mutual.

ATTRIBUTES

GPA; likes received / given (in- and out-degree); structural position (Katz-Bonacich); school vs. university.

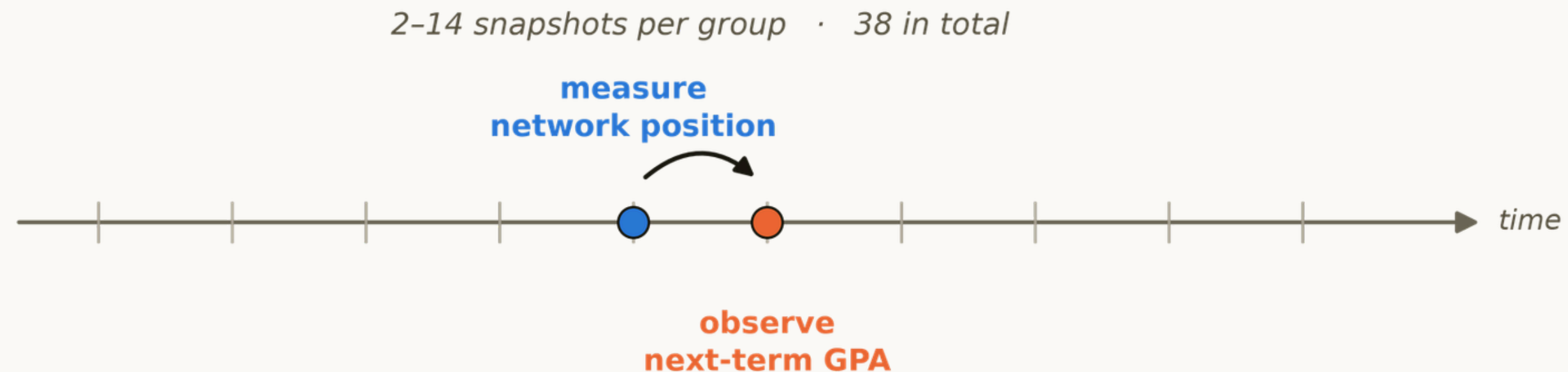


Same dataset as Smirnov & Thurner, network of Russian students

IDENTIFICATION

Which way does the arrow run?

Does network position lift grades, or do good grades just attract friends?



- Time order — position is measured before the GPA it predicts.
- Past GPA controlled — we compare students at the same current grade.
- Reverse arrow checked directly — after a GPA rise, do new friendships form?

So our claim is that position predicts later grades — not that it causes them.

PRIOR WORK

Smirnov & Thurner (2017): the effect is mostly selection

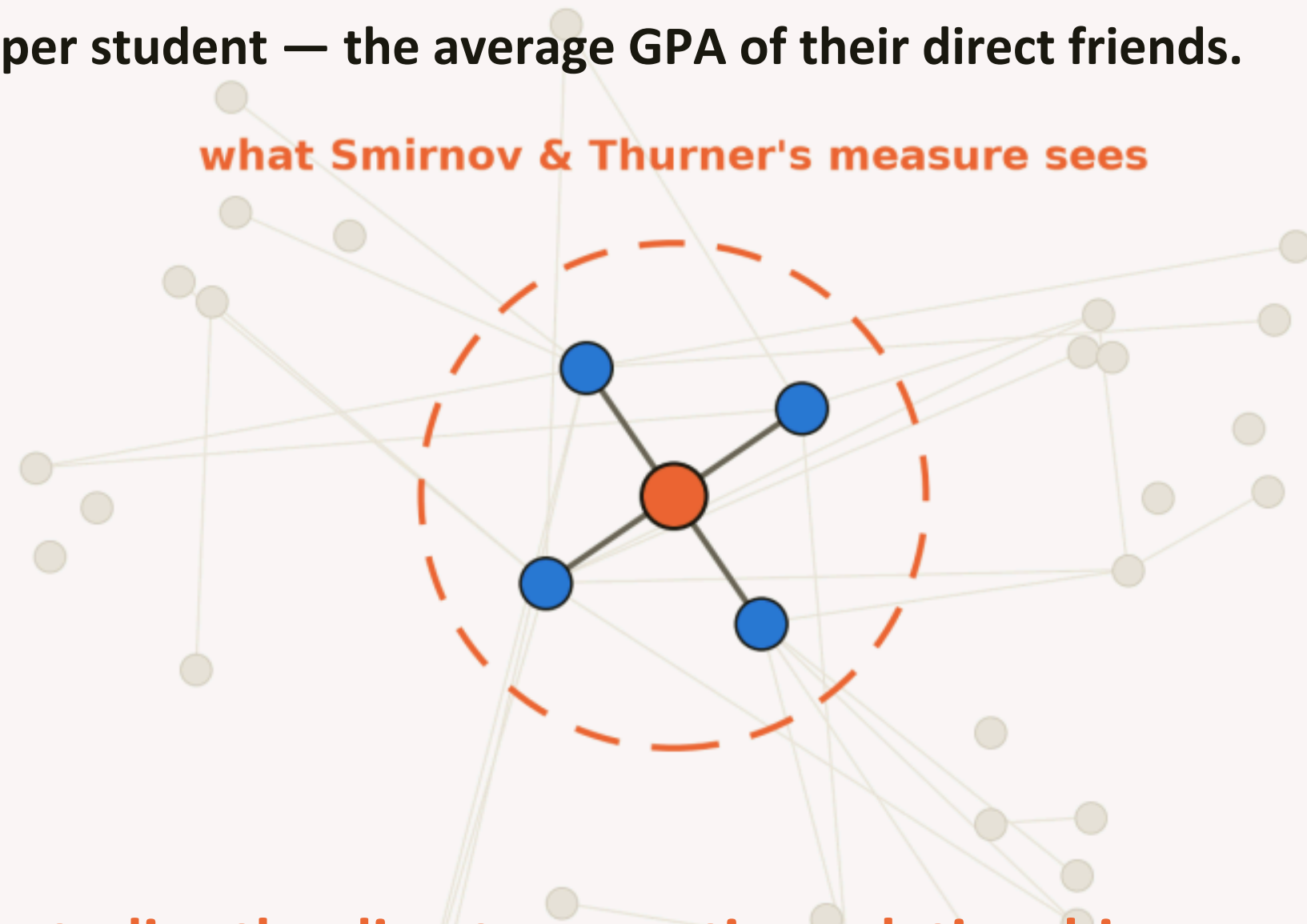
Students pick friends
with similar GPA

Friend groups end up
GPA-similar

GPA homophily

Their measure is one number per student — the average GPA of their direct friends.

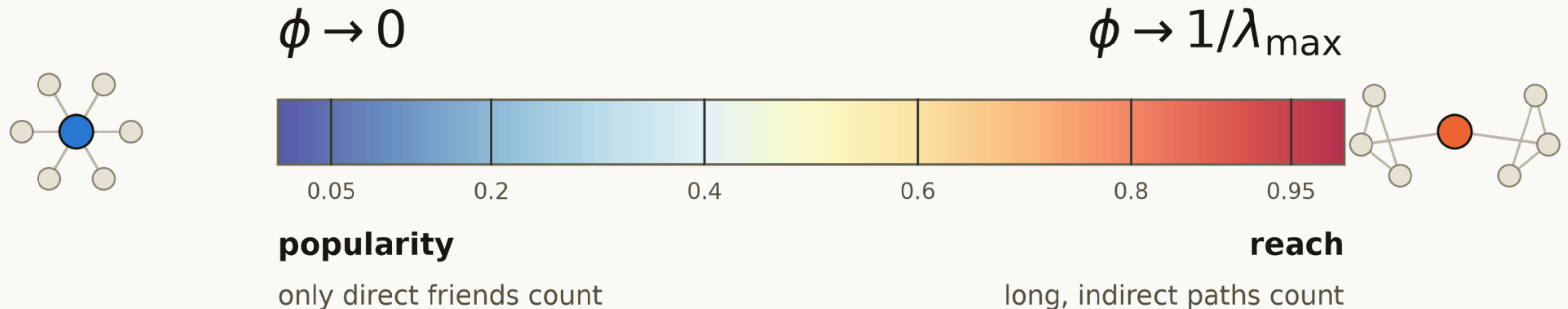
what Smirnov & Thurner's measure sees



→ It only studies the direct connection relationships.

Network position: more than counting friends

Katz–Bonacich centrality $b(g, \phi)$



Katz-Bonacich weights everyone you can reach — closer counts more.

Higher values of ϕ give more weight to deeper network positions in the centrality measure.

Predict next-term GPA from position — holding the rest fixed

$$\text{GPA}_{i, t+1} = \beta_0 + \beta_1 \text{Katz}_{i, t}(\phi) + \beta_2 \text{GPA}_{i, t} + \beta_3 \overline{\text{GPA}}_{i, t}^{\text{friends}} \\ + \beta_4 \text{InDeg}_{i, t} + \beta_5 \text{OutDeg}_{i, t} + \gamma_{\text{group}} + \varepsilon_{i, t}$$

Each control has a job:

- β_3 average friend GPA $\rightarrow \beta_1$ is net of friend-group composition
- β_4, β_5 raw in / out-degree $\rightarrow \beta_1$ is net of popularity

The real question: does centrality carry anything beyond simply counting friends?

What we expect — and what would prove us wrong

FROM THE MODEL

After controlling for direct connections (in-degree/out-degree), past GPA, and friends' GPA, we test whether deeper network position still has independent predictive power for future GPA, and whether this effect is stable across ϕ .

$$\beta_1 > 0$$

access to help lifts grades

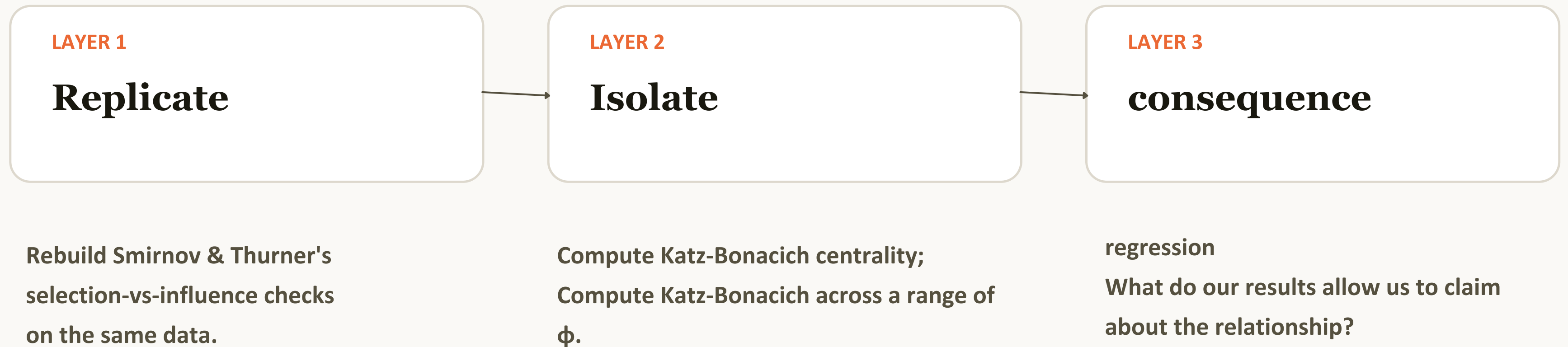
$$\beta_1 = 0$$

help and time-cost cancel

$$\beta_1 < 0$$

wide reach eats study time

Three layers of analysis



Thank you

- Data — Smirnov & Thurner (2017), Harvard Dataverse doi:10.7910/DVN/SZA9YW
- Model — Ballester, Calvó-Armengol & Zenou (2006); Calvó-Armengol, Patacchini & Zenou (2009)

